Sofya Dymchenko, Ph.D.

Grenoble, France | sofya.dymchenko@gmail.com | svdcvt.github.io

Summary

Applied researcher in **AI for Science and HPC**, specialising in data-driven surrogate modelling for scientific computing from data-centric and compute efficiency perspectives (**active learning, PINNs, distributed streamed training, foundation models**).

Work Experience

Doctoral Researcher, Grenoble, France — Mar 2022 – Apr 2026
Datamove team, INRIA and University of Grenoble-Alpes (UGA)

- Pioneered a research line on query-synthetic active learning for online training of deep PDE surrogates, contributing to the active sampling branch of Melissa, an open-source HPC&AI framework.
- Designed and implemented method Breed [2] that improves generalisation of a data-driven surrogate under limited computed budget by asynchronously curating data generation during training, experiments showing up to a magnitude lower maximum and average errors for 1D Kuramoto-Sivashinsky PDEs; orchestrated large-scale GPU training runs on Leonardo supercomputer.
- Proposed online generative active learning and contributed to diffusion-based OGAS method [1], which lowers high-percentile errors by up to 2x, outperforming pool-based baselines for transformer-based and neural operator surrogates on various 2D PDEs at 256^2 spatial resolution, e.g., Navier-Stokes.
- Introduced an active sampling method for PINNs training [3], which outperformed prior baselines by reducing training time by up tp 5x and lowering average errors by up to 2x for 1D Allen-Cahn PDE.
- Supervised 2 engineers and 2 master's students contributing to the Melissa framework, including project on online pretraining of foundation models for PDE surrogates.

Research Intern, Moscow, Russia — Jun 2021 – Dec 2021
Computational Intelligence Lab, Skolkovo Institute of Science and Technology (Skoltech)

- Extended master's thesis work on optimal transport to computational geometry and computer vision.

Research Intern, Moscow, Russia — Feb 2020 – Aug 2020
Media Algorithms Laboratory, Huawei R&D Center

- Trained generative attention models for personalized noise reduction in audio.

Education

PhD in Computer Science, Grenoble, France — Mar 2022 – Apr 2026
Laboratory of Informatics of Grenoble (LIG), UGA

- Thesis: High Performance Online Deep Neural Network Training from Synthetic Data with Active Learning for Scientific Computing. The manuscript is rated "Outstanding" (top 5%) by an external reviewer [link].

MSc in Applied Mathematics and Computer Science, Moscow, Russia — Sep 2019 – Jun 2021
Statistical Learning Theory program, Skoltech jointly with HSE (with honors, GPA: 4.89/5)

- Thesis: Developed modified optimal transport algorithm reducing complexity from quadratic to log-linear, applied to computational mesh generation [link].

BSc in Mathematics, Moscow, Russia — Sep 2015 – Jun 2019
Faculty of Mathematics, HSE (GPA: 7.87/10); Minor in Data Science, Faculty of Computer Science, HSE

Additional Experience & Skills

- Teaching (2019–2024): led Python course (3 years, 100+ students) incl. curriculum, TA supervision, lectures, "Best Course" award (2020); taught Data Science/ML seminars (HSE, UGA).
- Academic & community service: on-site organisation and outreach for Grenoble AI for Physical Sciences workshop [2024][2026]; reviewer for Workshop on Advancing Neural Network Training (NeurIPS'23, ICML'24); member of doctoral school student council (UGA), laboratory synergy (LIG); animal protection associations volunteer.

Technical skills: Python (PyTorch, JAX, Matplotlib, Pandas, Seaborn, Scikit-Learn, SciPy, NumPy, WandB), Git, Bash, SLURM, Linux/HPC environments, LaTeX, C++ (basic).

Languages: English (fluent), French (intermediate), Russian (native).

Publications

[1] P. Cesar, **S. Dymchenko**, A. Purandare, and B. Raffin. Learning Where to Simulate: Generative Active Sampling for Online PDE Surrogate Training. To appear in *the Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)* [preprint]

[2] **S. Dymchenko**, A. Purandare, and B. Raffin. MelissaDL x Breed: Towards Data-Efficient On-line Supervised Training of Multi-parametric Surrogates with Active Learning. In *SC-W 2024 - Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis*, Nov. 2024 [paper] [video]

[3] **S. Dymchenko** and B. Raffin. Loss-driven sampling within hard-to-learn areas for simulation-based neural network training. In *MLPS 2023 - Machine Learning and the Physical Sciences Workshop at NeurIPS 2023 - 37th Conference on Neural Information Processing Systems*, Dec. 2023 [paper]